

NAME: MUHAMMAD USMAN

REG. NO# 2024-EE-171

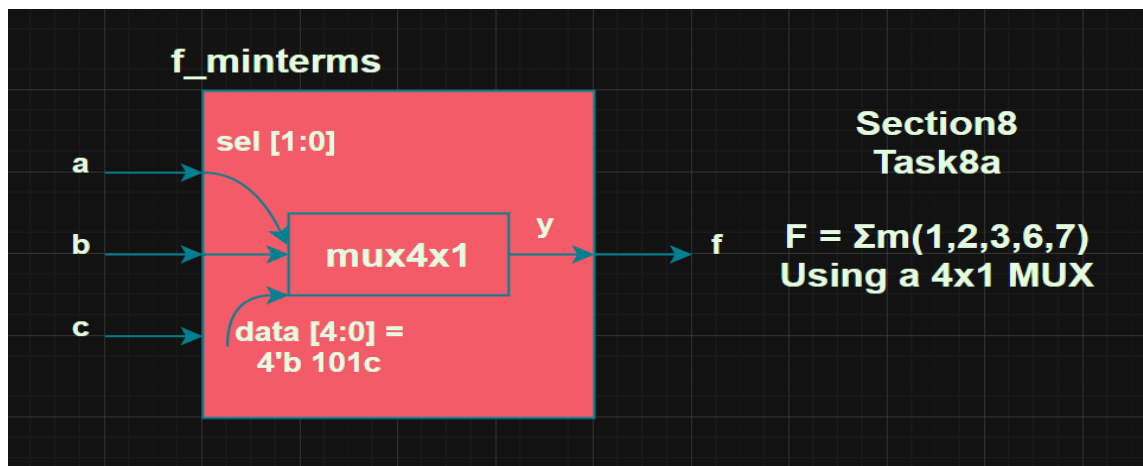
1-EDA Playground link of Task8a: <https://edaplayground.com/x/gHQr>

EDA Playground link of Task8b: <https://edaplayground.com/x/gHZy>

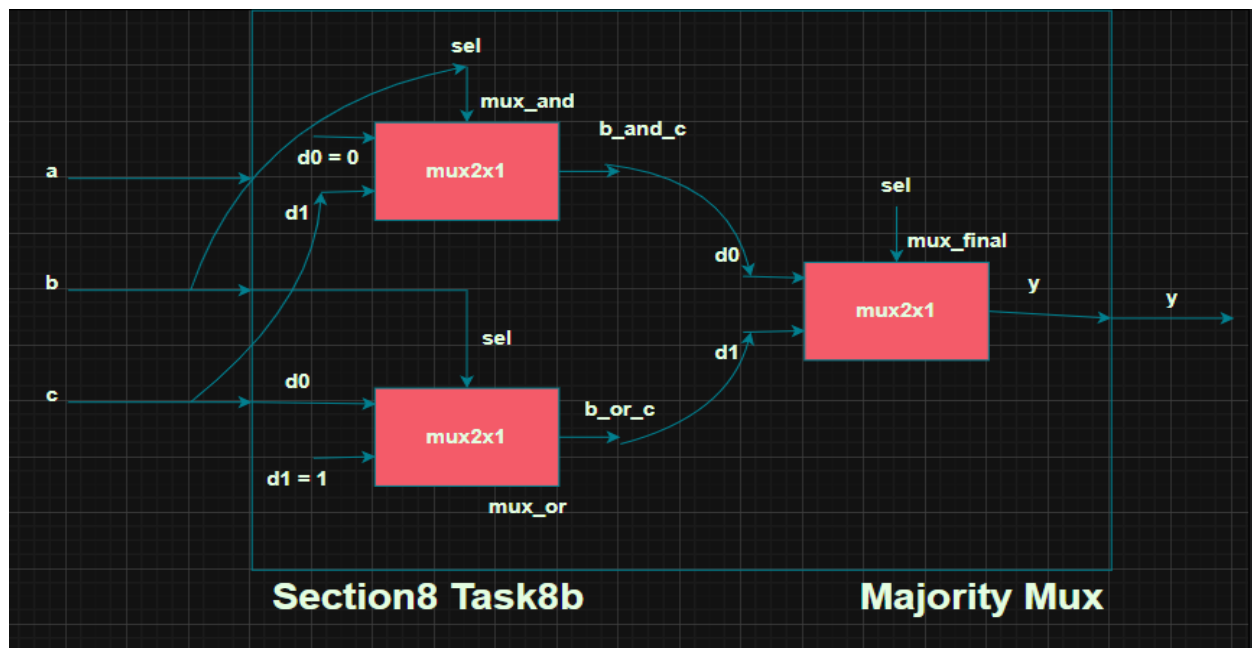
EDA Playground link of Task9a: <https://edaplayground.com/x/aBtK>

EDA Playground link of Task9b: <https://edaplayground.com/x/gHb2>

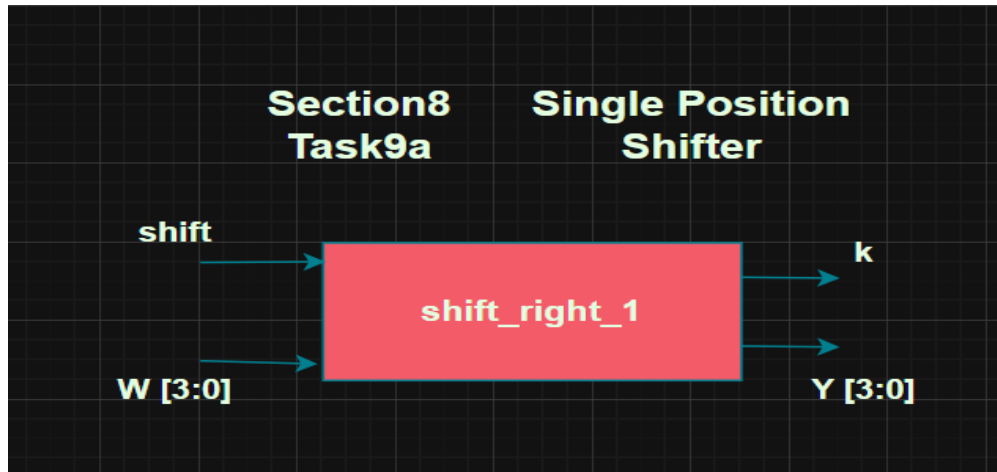
2-Schematic Diagram:



Task8a_schematic_diagram.png



Task8b_schematic_diagram.png



Task9a_schematic_diagram.png



Task9b_schematic_diagram.png

3-Output Screenshot:

```

Chronologic VCS simulator copyright 1991-2025
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full164; Runtime version X-2025.06-SP1_Full164; Jul 22 11:27 2026
[PASS]: a=0, b=0, c=0 -> f=0
[PASS]: a=0, b=0, c=1 -> f=1
[PASS]: a=0, b=1, c=0 -> f=1
[PASS]: a=0, b=1, c=1 -> f=1
[PASS]: a=1, b=0, c=0 -> f=0
[PASS]: a=1, b=0, c=1 -> f=0
[PASS]: a=1, b=1, c=0 -> f=1
[PASS]: a=1, b=1, c=1 -> f=1
$stop at time 80 Scope: tb File: testbench.sv Line: 38
uc1i% Finding VCD file...
./dump.vcd
[2026-07-22 15:27:05 UTC] Opening EPWave...
Done
  
```

Task8a_log_result.png

```

Chronologic VCS simulator copyright 1991-2025
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64; Jul 23 06:27 2026
[PASS]: a=0, b=0, c=0 -> y=0
[PASS]: a=0, b=0, c=1 -> y=0
[PASS]: a=0, b=1, c=0 -> y=0
[PASS]: a=0, b=1, c=1 -> y=1
[PASS]: a=1, b=0, c=0 -> y=0
[PASS]: a=1, b=0, c=1 -> y=1
[PASS]: a=1, b=1, c=0 -> y=1
[PASS]: a=1, b=1, c=1 -> y=1
$finish called from file "testbench.sv", line 31.
$finish at simulation time 80
VCS Simulation Report
Time: 80 ns
CPU Time: 0.470 seconds; Data structure size: 0.0Mb
Thu Jul 23 06:27:37 2026
Finding VCD file...
./dump.vcd
[2026-07-23 10:27:37 UTC] Opening EPWave...
Done

```

Task8b_log_result.png

```

Chronologic VCS simulator copyright 1991-2025
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64; Jul 22 11:27 2026
[PASS]: shift=0, w=0 -> Y=0, k=0
[PASS]: shift=0, w=1 -> Y=1, k=0
[PASS]: shift=0, w=10 -> Y=10, k=0
[PASS]: shift=0, w=11 -> Y=11, k=0
[PASS]: shift=0, w=100 -> Y=100, k=0
[PASS]: shift=0, w=101 -> Y=101, k=0
[PASS]: shift=0, w=110 -> Y=110, k=0
[PASS]: shift=0, w=111 -> Y=111, k=0
[PASS]: shift=0, w=1000 -> Y=1000, k=0
[PASS]: shift=0, w=1001 -> Y=1001, k=0
[PASS]: shift=0, w=1010 -> Y=1010, k=0
[PASS]: shift=0, w=1011 -> Y=1011, k=0
[PASS]: shift=0, w=1100 -> Y=1100, k=0
[PASS]: shift=0, w=1101 -> Y=1101, k=0
[PASS]: shift=0, w=1110 -> Y=1110, k=0
[PASS]: shift=0, w=1111 -> Y=1111, k=0
[PASS]: shift=1, w=0 -> Y=0, k=0
[PASS]: shift=1, w=1 -> Y=0, k=1
[PASS]: shift=1, w=10 -> Y=1, k=0
[PASS]: shift=1, w=11 -> Y=1, k=1
[PASS]: shift=1, w=100 -> Y=10, k=0
[PASS]: shift=1, w=101 -> Y=10, k=1
[PASS]: shift=1, w=110 -> Y=11, k=0
[PASS]: shift=1, w=111 -> Y=11, k=1
[PASS]: shift=1, w=1000 -> Y=100, k=0
[PASS]: shift=1, w=1001 -> Y=100, k=1
[PASS]: shift=1, w=1010 -> Y=101, k=0
[PASS]: shift=1, w=1011 -> Y=101, k=1
[PASS]: shift=1, w=1100 -> Y=110, k=0
[PASS]: shift=1, w=1101 -> Y=110, k=1
[PASS]: shift=1, w=1110 -> Y=111, k=0
[PASS]: shift=1, w=1111 -> Y=111, k=1
$stop at time 320 Scope: tb_shift_right_1 File: testbench.sv Line: 34
ucli% Finding VCD file...
./dump.vcd
[2026-07-22 15:27:49 UTC] Opening EPWave...
Done

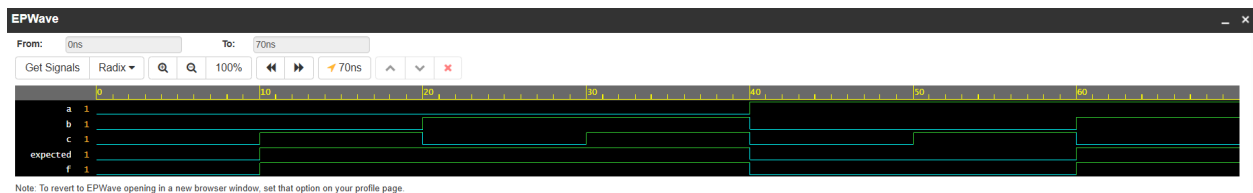
```

Task9a_log_result1.png

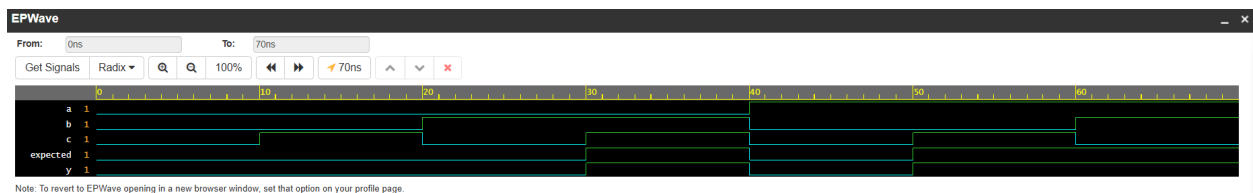
```
Chronologic VCS simulator copyright 1991-2025
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64; Jul 22 11:33 2026
[PASS]: sel=0 (0), W=1 -> Y=1
[PASS]: sel=0 (0), W=101 -> Y=101
[PASS]: sel=0 (0), W=1010 -> Y=1010
[PASS]: sel=0 (0), W=1111 -> Y=1111
[PASS]: sel=1 (1), W=1 -> Y=1000
[PASS]: sel=1 (1), W=101 -> Y=1010
[PASS]: sel=1 (1), W=1010 -> Y=101
[PASS]: sel=1 (1), W=1111 -> Y=1111
[PASS]: sel=10 (2), W=1 -> Y=100
[PASS]: sel=10 (2), W=101 -> Y=101
[PASS]: sel=10 (2), W=1010 -> Y=1010
[PASS]: sel=10 (2), W=1111 -> Y=1111
[PASS]: sel=11 (3), W=1 -> Y=10
[PASS]: sel=11 (3), W=101 -> Y=1010
[PASS]: sel=11 (3), W=1010 -> Y=101
[PASS]: sel=11 (3), W=1111 -> Y=1111
[PASS]: sel=0 (0), W=1 -> Y=1
[PASS]: sel=0 (0), W=101 -> Y=101
[PASS]: sel=0 (0), W=1010 -> Y=1010
[PASS]: sel=0 (0), W=1111 -> Y=1111
$stop at time 200 Scope: tb_barrel_shifter_4 File: testbench.sv Line: 43
uccli% Finding VCD file...
./dump.vcd
[2026-07-22 15:33:57 UTC] Opening EPWave...
Done
```

Task9b_log_result.png

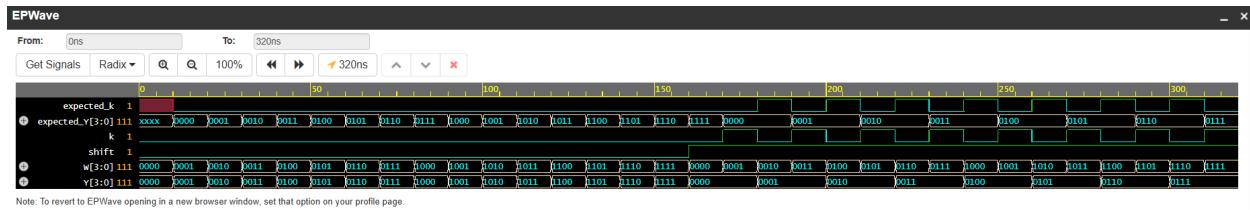
4-Waveform Screenshot:



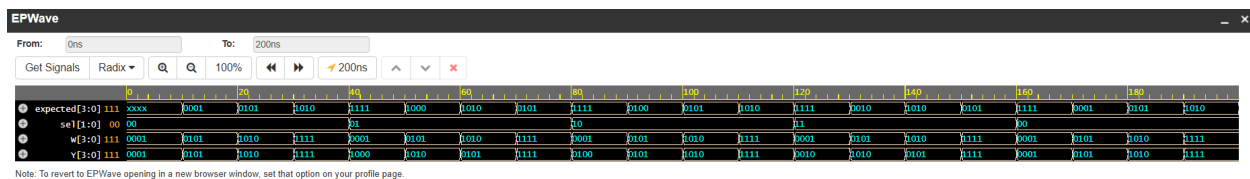
Task8a_waveform.png



Task8b_waveform.png



Task9a_waveform.png



Task9b_waveform.png

5-Explanation:

Explanation of “unique case”:

What “unique case” guarantees: It tells the simulator and synthesis tool that **exactly one** case item will match for any valid input combination (no overlaps, no missing cases).

Benefits:

1. The simulator issues a **warning** if multiple branches match (helps catch bugs).
2. The synthesizer can build a **parallel, faster** multiplexer instead of a priority chain.
3. It reduces the chance of unintended latches because the tool knows all cases are covered.

Without “unique case”: A plain “case” assumes no such guarantee; the tool builds more conservative logic, and overlapping/illegal “sel” values silently produce “X” or unexpected behavior.

In this barrel shifter, “sel” is a 2-bit signal, so exactly one of 00, 01, 10 and 11 matches at any time – perfect for unique case.

6-Truth Tables and MUX Implementation Derivation:

Truth Tables and MUX Implementation Derivation

can also be accessed using the following GitHub Link:

https://github.com/MuhammadUsman0001/MEDS_Training/blob/main/Module4%2FSection8%2FSection8.md

TASK 8: $F = \sum m(1,2,3,6,7)$ Using a 4x1 MUX

Step 1: Full Truth Table

Minterm	a	b	c	F
0	0	0	0	0
1	0	0	1	1
2	0	1	0	1
3	0	1	1	1
4	1	0	0	0
5	1	0	1	0
6	1	1	0	1
7	1	1	1	1

Step 2: MUX Implementation Derivation

Select lines: **a** (MSB) and **b** (LSB of select)

Group truth table rows by **a**, **b** :

a	b	Rows	F as function of c	Data Input
0	0	m0=0, m1=1	$F = c$	$d0 = c$
0	1	m2=1, m3=1	$F = 1$	$d1 = 1$
1	0	m4=0, m5=0	$F = 0$	$d2 = 0$
1	1	m6=1, m7=1	$F = 1$	$d3 = 1$

So, Data inputs: $\{d3, d2, d1, d0\} = \{1, 0, 1, c\}$

TASK 8 Extension: 3-Bit Majority Function Using Cascaded 2:1 MUXes

Truth Table (Majority: output = 1 when 2 or more inputs are 1)

a	b	c	Majority
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1